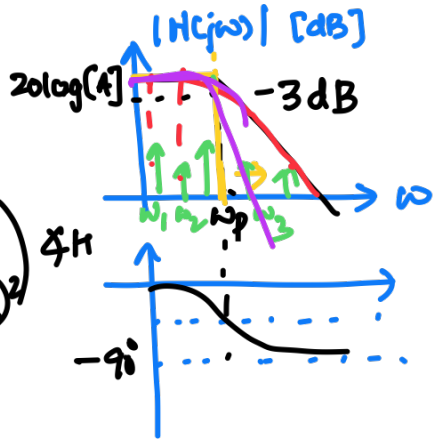
Lecture #27

Analog filters.

1. Low Pass filter (LPF)

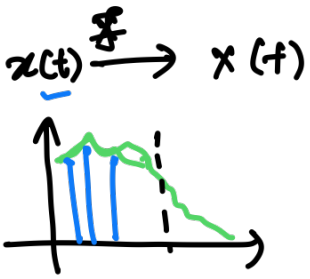


$$H = \frac{Y}{X} = \frac{A}{1+s/\omega_p}$$



$$20\log_{10}(|H|) = 20\log_{10}\left(\frac{A}{\sqrt{1+\left(\frac{\omega}{\omega_p}\right)^2}}\right)$$

$$\angle H(j\omega) = -\tan^{-1}\left(\frac{\omega}{\omega_p}\right)$$



$$x(t) = a_1\sin(\omega_1 t) + a_2\sin(\omega_2 t) + a_3\sin(\omega_3 t)$$

$$\text{Ideal } y(t) = \left(a_1\sin(\omega_1 t) + a_2\sin(\omega_2 t)\right)A$$

$$y(t) = |H(j\omega_1)|\, a_1\sin(\omega_1 t + \phi_1) + |H(j\omega_2)|\, a_2\sin(\omega_2 t + \phi_2) + \underbrace{|H(j\omega_3)|}_{\approx 0}\, a_3\sin(\omega_3 t + \phi_3)$$

$$\phi = -\tan^{-1}\left(\frac{\omega}{\omega_p}\right)$$

$$\omega_1 t + \phi_1 = \omega_1 t + \left(-\tan^{-1}\left(\frac{\omega_1}{\omega_p}\right)\right) = \omega_1\left\{t - \frac{1}{\omega_1}\tan^{-1}\left(\frac{\omega_1}{\omega_p}\right)\right\}$$

$t - t_{d_1}$

$$\omega_2 t + \phi_2 = \omega_2 t + \left(-\tan^{-1}\left(\frac{\omega_2}{\omega_p}\right)\right) = \omega_2\left\{t - \frac{1}{\omega_2}\tan^{-1}\left(\frac{\omega_2}{\omega_p}\right)\right\}$$

$t - t_{d_2}$

$$\tan^{-1}\left(\frac{\omega}{\omega_p}\right) \approx \frac{\omega}{\omega_p} \quad \times$$

- -3dB BW
- -1dB BW
- -0.1dB BW

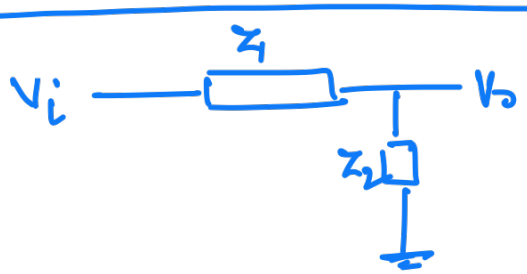
$$\frac{d}{d\omega}\left\{\frac{1}{\omega} \angle H(j\omega)\right\} = 0$$

"Group Delay"

$$|H(\omega_{-3dB})| = \frac{1}{\sqrt{2}}|H(0)| \checkmark$$

$$\omega_{BW} = \omega_{-3dB}$$

$$\frac{A}{\sqrt{1+\left(\frac{\omega_{-3dB}}{\omega_p}\right)^2}} = \frac{1}{\sqrt{2}} \times A \quad \Rightarrow \quad \omega_{-3dB} = \omega_p$$



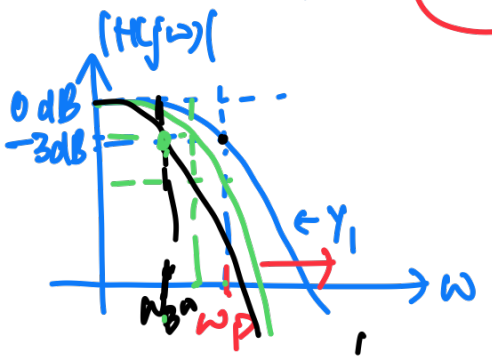
$$\frac{V_o}{V_i} = \frac{Z_2}{Z_1+Z_2} = \frac{1}{\frac{Z_1}{Z_2}+1}$$

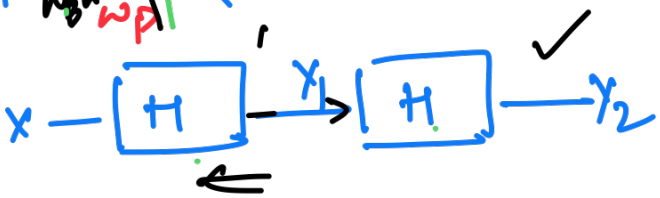
$Z_1 = R, \quad Z_2 = \frac{1}{sC} \quad \frac{V_o}{V_i} = \frac{1}{1+sRC}$

$Z_1 = sL, \quad Z_2 = R \quad \frac{V_o}{V_i} = \frac{1}{\frac{sL}{R}+1}$

} DC gain = 1

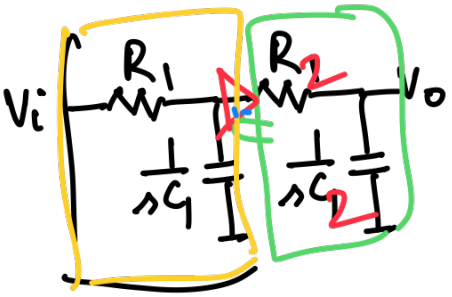
- Max. gain = 1

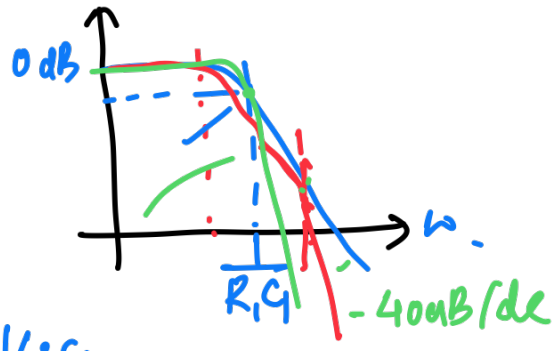




$$\frac{Y_2}{X} = H^2 \quad \Rightarrow \quad \left|\frac{Y_2}{X}\right| = |H|^2 = H_{total} = \frac{1}{1+s/\omega_p} \times \frac{1}{1+s/\omega_p}$$

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}$$





$$\frac{V_o}{V_i} = \frac{\left(R_1+\frac{1}{sC_1}\right) \| \frac{1}{sC_1}}{R_1 + \left(R_1+\frac{1}{sC_1}\right) \| \frac{1}{sC_1}} \times \frac{1/sC_1}{\frac{1}{sC_1}+R_1}$$

$$= \frac{\frac{1+sC_1R_1}{sC_1} \times \frac{1}{sC_1}}{\frac{1+sC_1R_1}{sC_1} + \frac{1}{sC_1}} \times \frac{1}{1+sC_1R_1}$$

$$R_1 + (\quad)$$

sL
$\frac{1}{sC}$

$$= \frac{(1+sC_1R_1)/sC_1(2+sC_1R_1)}{R_1 + \frac{1+sC_1R_1}{sC_1(2+sC_1R_1)}} \times \frac{1}{1+sC_1R_1}$$

$$= \frac{1+sC_1R_1}{2sC_1R_1 + s^2R_1^2C_1^2 + 1 + sC_1R_1} \times \frac{1}{1+sC_1R_1}$$

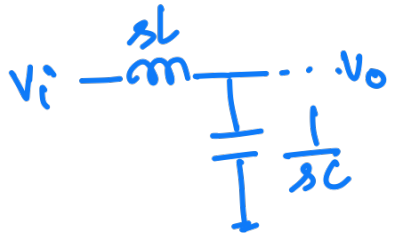
$$= \frac{1}{s^2R_1^2C_1^2 + 3sC_1R_1 + 1}$$

$$s = \frac{-3C_1R_1 \pm \sqrt{9C_1^2R_1^2 - 4R_1^2C_1^2}}{2R_1^2C_1^2}$$

$$= \frac{1}{RC}\left(-\frac{3}{2} \pm \frac{\sqrt{5}}{2}\right)$$

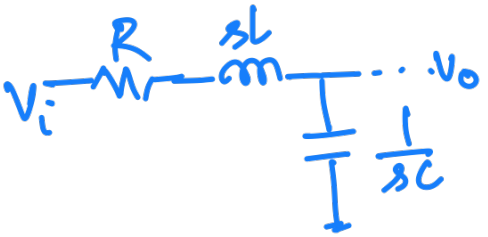
2.236
1.118
1.5

$$= \frac{1}{RC}\left(-2.618, -0.382\right)$$

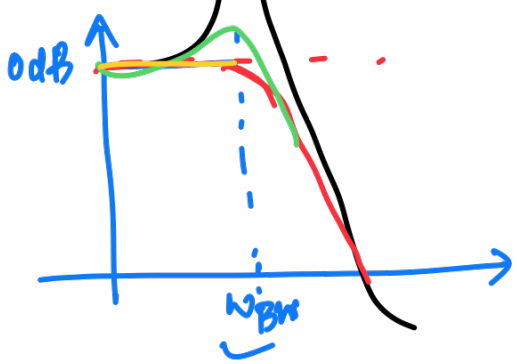


$$\frac{V_o}{V_i} = \frac{1}{s^2LC+1}$$

$$\left|\frac{V_o}{V_i}(j\omega)\right| = \left|\frac{1}{1-\omega^2LC}\right|$$



$$H = \frac{V_o}{V_i} = \frac{1}{s^2LC + sRC + 1} = \frac{1}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$



$$\frac{1/sC}{\frac{1}{sC} + R + sL}$$

$$\left|H(j\omega_{BW})\right| = \frac{1}{\sqrt{2}}\left|H(0)\right|$$